

7.4.32

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Question

$ABCD$ is a square of side length 2 units. C_1 is the circle touching all the sides of the square $ABCD$ and C_2 is the circumcircle of square $ABCD$. L is a fixed line in same plane and $\mathbf{R}$ is a fixed point.

- ① If $\mathbf{P}$ is any point of C_1 and $\mathbf{Q}$ is another point on C_2 , then

$$\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$$

- ① 0.75
- ② 1.25
- ③ 1
- ④ 0.5

- ① If a circle is such that it touches the line L and the circle C_1 externally, such that both the circles are on the same side of the line, then locus of centre of the circle
- ① ellipse
 - ② hyperbola
 - ③ parabola
 - ④ circle
- ② A L' through A is drawn parallel to BD . Point S moves such that its distances from the line BD and the vertex A are equal. If locus of S cuts L' at T_2 and T_3 and AC at T_1 , then area of $\triangle T_1 T_2 T_3$ is
- ① $1/2$ sq.units
 - ② $2/3$ sq.units
 - ③ 1 sq.units
 - ④ 2 sq.units

Let

Let:

The centre of incircle and circumcircle be **O**.

The radius of incircle be r_1 and that of circumcircle be r_2 .

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}; \mathbf{B} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (1)$$

$$\mathbf{C} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}; \mathbf{D} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (2)$$

Given:

$$r_1 = 1 \quad (3)$$

$$r_2 = \sqrt{2} \quad (4)$$

1st question

Let $\mathbf{P}$ be any point on incircle and $\mathbf{Q}$ be any point on circumcircle.

$\mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}$

$$\|\mathbf{X} - \mathbf{P}\|^2 = \|\mathbf{X}\|^2 + \|\mathbf{P}\|^2 - 2\mathbf{P} \cdot \mathbf{X} \quad (5)$$

(6)

Summation over all $\mathbf{X} | \mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}$:

$$\sum \|\mathbf{X} - \mathbf{P}\|^2 = \sum \|\mathbf{X}\|^2 + 4 \cdot \|\mathbf{P}\|^2 - 2\mathbf{P} \sum \mathbf{X} \quad (7)$$

(8)

1st question

For, $\mathbf{P} = \mathbf{P}$

$$\sum \|\mathbf{X} - \mathbf{P}\|^2 = \sum \|\mathbf{X}\|^2 + 4 \cdot \|\mathbf{P}\|^2 - 2\mathbf{P} \sum \mathbf{X} \quad (9)$$

$$4(1^2 + 1^2) + 4(1) - 2\mathbf{P} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right) \quad (10)$$

$$\therefore \|\mathbf{A} - \mathbf{P}\|^2 + \|\mathbf{B} - \mathbf{P}\|^2 + \|\mathbf{C} - \mathbf{P}\|^2 + \|\mathbf{D} - \mathbf{P}\|^2 = 12 \quad (11)$$

1st question

For, $\mathbf{P} = \mathbf{Q}$

$$\sum \|\mathbf{x} - \mathbf{Q}\|^2 = \sum \|\mathbf{x}\|^2 + 4\|\mathbf{Q}\|^2 - 2\mathbf{Q} \sum \mathbf{x} \quad (12)$$

$$4(1^2 + 1^2) + 4(2) - 2\mathbf{Q} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right) \quad (13)$$

$$\therefore \|\mathbf{A} - \mathbf{Q}\|^2 + \|\mathbf{B} - \mathbf{Q}\|^2 + \|\mathbf{C} - \mathbf{Q}\|^2 + \|\mathbf{D} - \mathbf{Q}\|^2 = 12 \quad (14)$$

1st question

conclusion:

$$\frac{12}{16} = 0.75 \quad (15)$$

Hence, option(a) is correct.

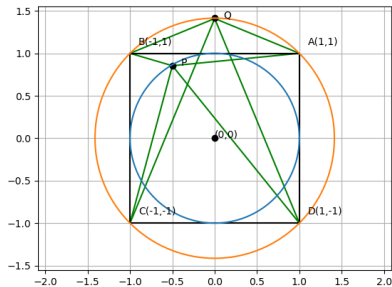


Figure: Plot of the given square and circles

2nd question

Let the radius of the moving circle be r , the centre of the circle be $\mathbf{X}$ and the line equation be $\hat{\mathbf{n}}^\top \mathbf{X} = c$,

2nd question

$$|\hat{\mathbf{n}}^\top \mathbf{X} - c| = r \quad (16)$$

$$||\mathbf{X}|| = r + 1 \quad (17)$$

$$||\mathbf{X}|| = |\hat{\mathbf{n}}\mathbf{X} - (c - 1)| \quad (18)$$

$$||\mathbf{X}||^2 = |\hat{\mathbf{n}}\mathbf{X} - (c - 1)|^2 \quad (19)$$

$$\mathbf{X}^\top \mathbf{X} = (\hat{\mathbf{n}}^\top \mathbf{X})^2 + (c - 1)^2 - 2\hat{\mathbf{n}}^\top \mathbf{X} (c - 1) \quad (20)$$

$$\mathbf{X}^\top \mathbf{X} - (\hat{\mathbf{n}}\mathbf{X})^2 + 2\hat{\mathbf{n}}^\top \mathbf{X} (c - 1) - (c - 1)^2 \quad (21)$$

$$\mathbf{X}^\top (I - \hat{\mathbf{n}}\hat{\mathbf{n}}^\top) \mathbf{X} + 2(c - 1)\hat{\mathbf{n}}^\top \mathbf{X} - (c - 1)^2 \quad (22)$$

2nd question

Equation (20) is the equation of parabola.
Hence, correct option is (c).

Plot

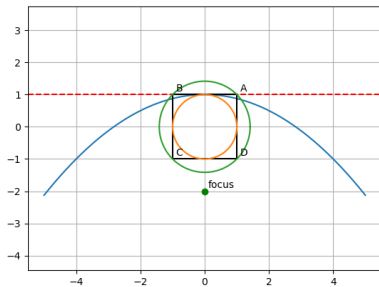


Figure: Plot of the given circles, square and locus of the point

3rd question

Let the point moving point be S and the line equation be $\mathbf{n}^\top \mathbf{S} = 0$.

$$\frac{|\mathbf{n}^\top \mathbf{S}|}{\|\mathbf{n}\|} = \|\mathbf{S} - \mathbf{A}\| \quad (23)$$

$$\frac{|\mathbf{n}^\top \mathbf{S}|^2}{\|\mathbf{n}\|^2} = \|\mathbf{S} - \mathbf{A}\|^2 \quad (24)$$

$$\frac{|\mathbf{S}^\top \mathbf{n} \mathbf{n}^\top \mathbf{S}|}{\|\mathbf{n}\|} = (\mathbf{S} - \mathbf{A})^\top (\mathbf{S} - \mathbf{A}) \quad (25)$$

$$\mathbf{S}^\top \left(\mathbf{I} - \hat{\mathbf{n}} \hat{\mathbf{n}}^\top \right) \mathbf{S} - 2\mathbf{A}^\top \mathbf{S} + \mathbf{A}^\top \mathbf{A} = 0 \quad (26)$$

3rd question

Equation 24 is the locus of the moving point which is a parabola.

Let: $\mathbf{m}_1$ be the direction vector of line AC, $\mathbf{m}_2$ be the direction vector of line L' .

$$\mathbf{m}_1 = \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (27)$$

$$\mathbf{m}_2 = \begin{pmatrix} 2 \\ -2 \end{pmatrix} \quad (28)$$

3rd question

The equation of line L'

$$\mathbf{S} = \mathbf{A} + t\mathbf{m}_2 \quad (29)$$

The equation of the line AC

$$\mathbf{S} = \lambda\mathbf{m}_1 \quad (30)$$

Substituting equation (30) in (24) we get $\lambda = \frac{1}{2}$:

$$\mathbf{T}_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (31)$$

3rd question

Substituting (29) in (24) we get $t = \frac{-1}{2}, \frac{1}{2}$

$$\mathbf{T}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad (32)$$

$$\mathbf{T}_3 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad (33)$$

Now, finding area of the triangle:

$$\Delta T_1 T_2 T_3 = \frac{1}{2} \left\| \begin{pmatrix} \mathbf{T}_2 - \mathbf{T}_1 & \mathbf{T}_3 - \mathbf{T}_2 \end{pmatrix} \right\| \quad (34)$$

$$\Delta T_1 T_2 T_3 = 1 \quad (35)$$

Option (c) is correct.

Plot

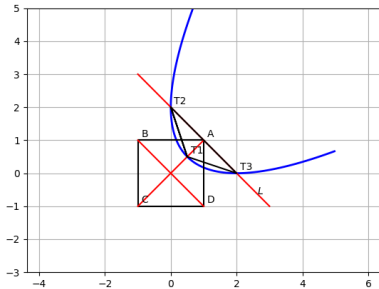


Figure: Plot of the given circles, square and locus of the point

```
#include <stdio.h>
#include <math.h>
void get_results(double *out_data) {
    double side = 2.0;
    double r1 = side / 2.0;
    double r2 = side / sqrt(2.0);
    double ratio = (r1 * r1 + r1 * r1 + r1 * r1 + r1 * r1) /
                   (r2 * r2 + r2 * r2 + r2 * r2 + r2 * r2);
    double k = 2.0;
    double focus_x = 0.0;
    double focus_y = k + r1;
    double directrix_y = k - r1;
    double area = 2.0 / 3.0;
```

C code

```
double a = 1;
double b = -1;
out_data[0] = ratio;
out_data[1] = focus_x;
out_data[2] = focus_y;
out_data[3] = directrix_y;
out_data[4] = area;
out_data[5] = a; //Ax
out_data[6] = a; //Ay
out_data[7] = b; //Bx
out_data[8] = a; //By
out_data[9] = b; //Cx
out_data[10] = b; //Cy
out_data[11] = a; //Dx
out_data[12] = b; //Dy
out_data[13] = 0.5; //T1x
out_data[14] = 0.5; //T1y;
out_data[15] = 0; //T2x;
out_data[16] = 2; //T2y
```

```
import ctypes as ct
import numpy as np
def get_results():
    lib = ct.CDLL("./problem.so")
    lib.get_results.argtypes = [ct.POINTER(ct.c_double)]
    data_type = ct.c_double * 19
    data = data_type()
    lib.get_results(data)
    arr = np.array(list(data))
    ratio, focus_x, focus_y, directrix_y, area, Ax, Ay, Bx, By,
        Cx, Cy, Dx, Dy, T1x, T1y, T2x, T2y, T3x, T3y= arr
    return ratio, np.array([focus_x, focus_y]), directrix_y, area
        , Ax, Ay, Bx, By, Cx, Cy, Dx, Dy, T1x, T1y, T2x, T2y, T3x
        , T3y
```

```
import numpy as np
import matplotlib.pyplot as plt
from call import get_results
ratio, focus, directrix_y, area, Ax, Ay, Bx, By, Cx, Cy, Dx, Dy,
    T1x, T1y, T2x, T2y, T3x, T3y = get_results()
theta = np.linspace(0, 2*np.pi, 200)
A = ([Ax, Bx, Cx, Dx, Ax, Bx])
B = ([Ay, By, Cy, Dy, Ay, By])
C = ([Ax, 0])
D = ([Ay, 1.41])
plt.plot(C, D, color='green')
C = ([Bx, 0])
D = ([By, 1.41])
plt.plot(C, D, color='green')
```

```
C = ([Cx, 0])
D = ([Cy, 1.41])
plt.plot(C, D, color='green')
C = ([Dx, 0])
D = ([Dy, 1.41])
plt.plot(C, D, color='green')
C = ([Ax, -0.5])
D = ([Ay, 0.854])
plt.plot(C, D, color='green')
C = ([Bx, -0.5])
D = ([By, 0.854])
plt.plot(C, D, color='green')
C = ([Cx, -0.5])
D = ([Cy, 0.854])
plt.plot(C, D, color='green')
C = ([Dx, -0.5])
D = ([Dy, 0.854])
plt.plot(C, D, color='green')
```

```
plt.plot(A, B, color='black')
plt.plot(0, 0, "ko")
plt.plot(0, 1.414, "ko")
plt.text(Ax+0.1, Ay+0.1, "A(1,1)", fontsize = 10, color = 'black'
)
plt.text(Bx+0.1, By+0.1, "B(-1,1)", fontsize = 10, color = 'black'
')
plt.text(Cx+0.1, Cy+0.1, "C(-1,-1)", fontsize = 10, color = '
black')
plt.text(Dx+0.1, Dy+0.1, "D(1,-1)", fontsize = 10, color = 'black'
')
plt.text(0,0, "(0,0)", fontsize = 10, color = 'black')
plt.text(0.1, 1.414, "Q", fontsize = 10, color = 'black')
plt.plot(-0.5, 0.854, "ko")
plt.text(-0.4, 0.854, "P", fontsize = 10, color = 'black')
```

```
plt.plot(np.cos(theta), np.sin(theta), label="Inner Circle C1")
plt.plot(np.sqrt(2)*np.cos(theta), np.sqrt(2)*np.sin(theta),
         label="Outer Circle C2")

plt.axis("equal")
plt.grid(True)
plt.savefig("../figs/plot1.png")
plt.show()
```

```
import numpy as np
import matplotlib.pyplot as plt
from call import get_results

ratio, focus, directrix_y, area, Ax, Ay, Bx, By, Cx, Cy, Dx, Dy,
    T1x, T1y, T2x, T2y, T3x, T3y = get_results()
A = ([Ax, Bx, Cx, Dx, Ax, Bx])
B = ([Ay, By, Cy, Dy, Ay, By])

x = np.linspace(-5, 5, 300)

p = (focus[1] - directrix_y)
y = -(x**2) / (4*p)+1
```

```
plt.plot(x, y, label="Locus (Parabola)")
plt.axhline(directrix_y, color='r', linestyle='--', label='
    Directrix')
plt.plot(focus[0], -focus[1]+1, 'go')
plt.plot(A, B, color='black')
plt.text(Ax+0.1, Ay+0.1, "A", fontsize = 10, color = 'black')
plt.text(Bx+0.1, By+0.1, "B", fontsize = 10, color = 'black')
plt.text(Cx+0.1, Cy+0.1, "C", fontsize = 10, color = 'black')
plt.text(Dx+0.1, Dy+0.1, "D", fontsize = 10, color = 'black')
plt.text(focus[0]+0.1, -focus[1]+1+0.1, 'focus', fontsize = 10,
    color = 'black')
```

```
theta = np.linspace(0, 2*np.pi, 200)
plt.plot(np.cos(theta), np.sin(theta), label="Inner Circle C1")
plt.plot(np.sqrt(2)*np.cos(theta), np.sqrt(2)*np.sin(theta),
         label="Outer Circle C2")
plt.axis("equal")
plt.grid(True)
plt.savefig("../figs/plot2.png")
plt.show()
```

```
import numpy as np
import matplotlib.pyplot as plt
from call import get_results
ratio, focus, directrix_y, area, Ax, Ay, Bx, By, Cx, Cy, Dx, Dy,
    T1x, T1y, T2x, T2y, T3x, T3y = get_results()
A = ([Ax, Bx, Cx, Dx, Ax, Bx])
B = ([Ay, By, Cy, Dy, Ay, By])
C = ([T1x, T2x, T3x, T1x, T2x])
D = ([T1y, T2y, T3y, T1y, T2y])
def Q(x, y):
    return x**2+y**2-2*x*y-4*x-4*y+4
E = ([Bx, Dx])
F = ([By, Dy])
```

```
X = np.linspace(-1, 3, 100)
Y = -X + 2

x_range = np.linspace(-3, 5, 400)
y_range = np.linspace(-3, 5, 400)
S, P = np.meshgrid(x_range, y_range)
Z = Q(S, P)
parabola = plt.contour(S, P, Z, levels=[0], colors='blue',
                        linewidths = 2)
```

```
triangle = np.array([[0,0], [2,0], [1, np.sqrt(3)/2]])  
plt.plot(A, B, color='black')  
plt.plot(E, F, "r-")  
plt.plot(X, Y, "r-")  
  
E = ([Ax, Cx])  
F = ([Ay, Cy])  
plt.plot(E, F, "r-")
```

```
plt.text(Ax+0.1, Ay+0.1, "A", fontsize = 10, color = 'black')
plt.text(Bx+0.1, By+0.1, "B", fontsize = 10, color = 'black')
plt.text(Cx+0.1, Cy+0.1, "C", fontsize = 10, color = 'black')
plt.text(Dx+0.1, Dy+0.1, "D", fontsize = 10, color = 'black')
plt.plot(C, D, color='black')
plt.text(T1x+0.1, T1y+0.1, "T1", fontsize = 10, color = 'black')
plt.text(T2x+0.1, T2y+0.1, "T2", fontsize = 10, color = 'black')
plt.text(T3x+0.1, T3y+0.1, "T3", fontsize = 10, color = 'black')
plt.text(2.62, -0.64, r'$L^-$')
```

```
plt.axis("equal")  
plt.grid(True)  
plt.savefig("../figs/plot3.png")  
plt.show()
```